

Assignment Introduction

You are tasked with designing a **web-based communication system** to support university students doing **group work** (e.g., assignments and projects). Your design should go *beyond existing tools* like Slack, Discord, or Teams by focusing on the unique needs of students and their group work.

In this project, you will follow a *User-Centered Design (UCD)* process to **identify user requirements**, **design alternatives**, and **develop lo-fi prototypes**. Then, you will use a selected set of Large Language Models (LLMs) to assist in **developing hi-fi prototypes**.

Considering real world deployment, you will define **security requirements** for your system. You will **evaluate vulnerabilities** in the generated systems and apply **security solutions** to address them.

Finally, you will create an **evaluation plan** to see how they meet the design and user requirements.

Things to note:

- The project is divided into three (3) phases, and each phase has a specific assignment that you have to submit on Canvas. Phases are designed so that you can chronologically follow them with overlaps.
- This is a group project. There should be 4 students per group, and where an exception can be made, groups of 3 students are allowed. You must make a case to your tutor why it has to be a group of 3. All the group members must be from the same tutorial group.
- The groups will be formed in the W01 tutorial with the students present during the tutorial. Once you form the group, inform your tutor; they will set up your group in Canvas.
- We highly encourage you to use modern development tools (e.g., AI-assisted IDEs like Cursor or Windsurf) to minimise overhead, allowing you to focus on usability and security concepts.
- Keep all of your answers concise and on-point, and do not repeat yourself. We highly encourage you to develop the report concurrently while working on the project tasks, rather than starting the report after all the tasks are completed.
- The submission must be made in PDF format. All submitted *text* (report, codes, etc.) must be in *text* format that is readable by [Turnitin](#). Do not convert the text part of the report to images (e.g., jpeg). If you submit in such a format, we may ask you to resubmit the report, and you may be subject to late submission penalties. Please read the **Adobe PDF** section of the [Turnitin file requirements](#) section to see how you can ensure it is readable. The images in your report (e.g., images of your prototype) do not need to meet this requirement. $\text{\LaTeX}$ generated PDF files naturally meet this requirement unless you further process them.
- Every project member should contribute equally to the project (both the project tasks and writing the report) and **individual contributions should be reflected using the diary section of the report (Page 11)**. All the project-related code should be managed through the [USyd Github](#).
- **Report Template:** The template for the report will be provided in Canvas.
- **All the project members' names, SIDs, the project group name (as it appears in Canvas) should be included as the author information.** Incorrect template, missing author or group information will incur penalties (-10% per phase).

Phase 1 - Project Plan (50 pts [weight 5%])

In this phase, you are required to develop the project plan and the scope. This involves several tasks, as outlined below. Each task should be recorded in the report following the template. Please follow the word counts per task where they apply. Also, include the *diary* for the phase as outlined on Page 11 at the end of the report.

Tasks:

1. Formulate a preliminary system description for the proposed system. It must include:- **Task total 9pts**
 - (a) The goals and aims of the system (less than 50 words).
 - (b) Ideate at least 1 feature you could integrate into the proposed system that current communication platforms do not have (less than 30 words).
 - (c) State how the above feature will be helpful for students to do their group work (less than 30 words).
2. Selecting potential implementation mechanisms. **- Task total 41pts**
 - (a) Survey and list 6 (4 for 3 student groups) different generative AI platforms (with URLs to their website) you can use to help implement the system above. They should not be different versions of the same platform. You should not use *DeepSeek* as the university suggests you not to. We encouraged you to use freely available versions.
 - (b) For each GenAI platform above, list one potential advantage and one disadvantage of using them to help implement the proposed system. This can be based on the known capabilities and issues of the selected platform.
 - (c) Write a prompt that you imagine will help you to develop a simple prototype for the web interface with basic functionality for a quick test of the proposed system.
 - (d) Using the same prompt above on each GenAI platform above, generate different prototypes for the proposed system. There should be 6 (4 for 3 student groups) prototypes generated per group. Try them out and add screenshots of the developed prototype system (1 to 2 screenshots per platform is sufficient) showing key features to your report.
 - (e) Examine the design and the functionality of the generated interface, and rate each prototype generated by the GenAI platforms, and describe the reasoning for your rating (less than 50 words per platform).

Phase 2 - Requirement Gathering & Design (100pts [weight 10%])

In this phase, you will gather user requirements, develop conceptual designs and prototypes for your group project. This involves several tasks, as outlined below. Each task will either have a reporting or a live demo during the tutorial. Also, include the *diary* for the period as outlined on Page 11 at the end of the report.

Tasks:

1. Identify user requirements for the project. **- Task total 40pts**
 - (a) Develop a plan for a semi-structured interview to gather requirements for the proposed project. The plan must include: **- 10pts**
 - The goal and a brief description of the interview.
 - Who the potential participants are, along with the inclusion/exclusion criteria.
 - Plans for pilot studies and ethical considerations (including required ethics documents)¹.
 - Planned activities, interview questions, data to be collected, data recording formats, and analysis plan.
 - (b) Recruit participants and conduct the interview. Include transcripts in the appendix. Groups of 4 must interview 4 participants, where groups of 3 - 3 participants. Participants must be recruited outside the INFO2222 (2026) cohort. **- 10pts**
 - (c) Conduct a thematic analysis. Document the process followed and the results (themes). **- 10pts**
 - (d) Using the themes as references, create a table of features required for your new system along with a short justification for each. Clearly state if the selected features are the same, refined from or new compared to the features you came up with in Phase 1. Compare these to existing systems (Teams, Slack, etc.), and clearly outline if each feature exists in them. We recommend you find at least 1-2 unique features matching the context of the project scope. **- 5pts**
 - (e) Write a *user scenario* of how your proposed project will be used. Please read these articles to learn about how to write a *user scenario*: [[article 1](#), [article 2](#)] **- 5pts**
2. Designing alternatives. **- Task total 20pts**
 - (a) Follow the “10 plus 10 method” to sketch your ideas for a potential web-based user interface to implement the system. As a group, select 2 alternative designs. Please briefly explain (max 200 words per design) your two ideas in the report (include your sketches from the selected designs, limited to 6 sketches per design). **- 10pts**
 - (b) Either choose between (and/or) merge the two designs into one design solution and justify it with your reasoning. Develop a paper prototype (vertical + horizontal) for your chosen design. For the vertical prototype, we recommend you focus on one key feature. Include photos of your paper prototype in the report. Paper prototypes should be hand-drawn (you may use a digital device to draw if preferred), and you must have physical paper prototypes (e.g., printed) to be shown during the tutorial. **- 10pts**
3. Hi-Fi prototypes. **- Task total 10pts**
 - (a) Using either an AI tool you explored and found to be useful in Phase 1, or using a prototyping tool like Figma, Stitch, etc., develop a high-fidelity prototype. The following considerations should be taken into account: **- 10pts**

¹Please check the [HREC](#) guidelines of the university, for this study, you will not need the ethics approval since we will not publish the results. But you should download the template for consent and information sheets, and include those into your study process, i.e., collect informed consent.

- Use the paper prototype as the basis for your hi-fi prototype. You can use the aspects of it to generate prompts, or even directly use them as input screenshots to generate the hi-fi prototype.
 - We recommend following technologies (you can integrate those into your prompts), but you don't have to limit yourself for these. Implementation choices are yours. These are very popular technologies that may have a lot of resources available online. And also, if you use these technologies, we will be able to help you with some technical queries along the way.
 - Next.js: A framework for both the frontend (UI) and the backend server logic. Best to use **App Router** and **TypeScript**.
 - Tailwind CSS: An easy to use CSS framework to design responsive web interfaces.
 - Socket.io: A powerful JavaScript library for real-time, bidirectional, event-based communication (for the chat). You can use it for features like instant message broadcasting, online/offline status, and notifications.
 - You may also consider using a simple local database, e.g., **SQLite**, **Local MongoDB**, or **RxDB**.
 - While the above are recommendations, you may choose the appropriate technologies based on experience and knowledge of the team members. For instance, if you are familiar with Python, technologies like **Flask** or **Django** may work better.
 - You must clearly document all prompts and AI-generated content produced using AI-assisted tools as an appendix in your report, for reproducibility and to address plagiarism concerns. Any file, function, or substantial code snippet created or modified by AI assistance must be explicitly listed (see Table 1 for an example).
 - Please provide your source files (USyd Github link in the report) in your final submission with documentation (in a **README.md**).
 - Make sure the system has some functionality (it is OK if not all the functions are working) at the live demo during the tutorial (Week 06). You must clearly indicate which functions do not work in the report.
- (b) Live demo during tutorial, showcase your themes, lo-fi and hi-fi prototypes (5 mins per group, followed by 5 mins Q&A). The attendance during the demo and Q&A is needed to gain the points for this part of the assessment.
- Task total 30pts**

File Path	Description	AI Tool	Prompt Used
hello.py, greeting.py	Scripts printing greeting messages	claude-3.7-sonnet	"Write Python scripts that greet the user differently based on the time of day."
README.md	Basic project documentation	gpt-4o-mini	"Generate a short README file describing Python scripts that greet users based on the current time."

Table 1: Summary of AI-generated files and prompts used

Phase 3 - Security, Evaluation & Report (150pts [15%])

You will enhance the security of the group communication system built in Phase 2, demonstrate the impact of missing security measures, and evaluate the security awareness of Large Language Models (LLMs). The goal is to deepen your understanding of security vulnerabilities and defenses through hands-on implementation and analysis. You are encouraged to use modern development tools (e.g., AI-assisted IDEs like Cursor) to minimize overhead, allowing you to **focus on security features and concepts**.

Key objectives:

- *Task 1 (45 points)*: Implement critical security features (e.g., authentication, encryption, secure transmission, certificate validation) and ensure their correct functionality.
- *Task 2 (30 points)*: Conduct controlled experiments to demonstrate vulnerabilities arising from missing or misconfigured security measures, illustrating worst-case attack scenarios (e.g., man-in-the-middle, replay attacks, weak password storage).
- *Task 3 (15 points)*: Conduct the Think Aloud study to assess usability of your system.
- *Task 4 (60 points)*: Assess and compare how different LLMs handle security-related queries, identifying their strengths, limitations, and potential biases in security advice.

Due: 15th of May

Submission List:

1. 5-minute video for Task 1
2. 3-minute video per security vulnerability for Task 2
3. **Each student must present part of the presentation in either Task 1 or Task 2; any student who is absent or presents for less than 2 minutes will receive reduced individual marks on a pro rata basis.**
4. One consolidated report covering: (i) the think-aloud study, and (ii) LLM security. No word limit – concise, scientific writing is preferred (Less is more).
5. Weekly diary
6. LLM prompt documentation
7. Source files (via Git repository)

Task 1 - Security Implementation**- Task total 45 pts**

Enhance the communication system you developed in Phase 1, so that it ensures user authentication, message confidentiality, and integrity.

Submission Requirement:

- A **5-minute video** demonstrating your implementation.
- You must clearly document all prompts and AI-generated content produced using AI-assisted tools as an appendix in your report, for reproducibility and to address plagiarism concerns. Any file, function, or substantial code snippet created or modified by AI assistance must be explicitly listed (see Table 1 for an example).
- Please provide your source files (USyd Github link in the report) in your final submission.

Grading Criteria (based on recorded video and short Q&A in week 12's tutorial):**1. Secure Password Storage**

- (a) Explain the characteristics of a strong hashing algorithm and justify your choice. **- 5pts**
- (b) Implement a secure password hashing method with proper salting. Plaintext storage or weak hashing will result in deductions. **- 10pts**

2. Server Authentication on Login

- (a) The client must verify the server's certificate before transmitting credentials. **- 5pts**
- (b) If a hardcoded CA public key is used, discuss its security implications. Explain how the server certificate is generated (including the CA certificate). **- 5pts**

3. Secure Password Transmission

- (a) Passwords must be transmitted over a secure channel (e.g., TLS 1.2+). Plaintext transmission of passwords is strictly prohibited. **- 10pts**

4. Secure Message Transmission

- (a) Messages must be encrypted end-to-end (E2EE). **- 10pts**

Task 2 - Security Demonstrations**- Task total 30 pts**

Demonstrate at least **two distinct security vulnerabilities** by disabling or misconfiguring security features in your system. The goal is to illustrate the consequences of neglecting security best practices.

Submission Requirement:

- A **3-minute video** per vulnerability, demonstrating the attack. Clearly explain the attack, its impact, and how proper security measures mitigate it.
- You must clearly document all prompts and AI-generated content produced using AI-assisted tools as an appendix in your report, for reproducibility and to address plagiarism concerns. Any file, function, or substantial code snippet created or modified by AI assistance must be explicitly listed (see Table 1 for an example).
- Please provide your source files (USyd Github link in the report) in your final submission.

Grading Criteria (based on recorded video and short Q&A in week 12's tutorial):**1. Vulnerability 1 Demonstration**

- Present an attack scenario caused by disabling or misconfiguring a security feature. **- 10pts**
- Explain why the vulnerability occurs and how to fix it. **- 5pts**

2. Vulnerability 2 Demonstration

- Demonstrate a second, distinct attack scenario. **- 10pts**
- Explain why the vulnerability occurs and how to fix it. **- 5pts**

Example Attack Scenarios (Choose at least one):

- **Message Interception:** Disable encryption or certificate validation to demonstrate eavesdropping via a network sniffer.
- **Weak Password Storage:** Store passwords in plaintext or use weak hashing (e.g., MD5, SHA-1) and demonstrate password retrieval by an attacker.
- **SQL Injection:** Exploit improper input handling to manipulate database queries.
- **Denial-of-Service (DoS):** Remove rate limiting and show how repeated requests degrade system performance.

Task 3 - User Evaluation**- Task total 15pts**

1. Develop a study plan for evaluating the Hi-Fi prototypes for usability using the [Think Aloud](#) methodology. The following considerations should be taken into account: **- 5pts**
 - The goal and a brief description of the study.
 - Who the potential participants are, along with inclusion/exclusion criteria. You can recruit the participants from other groups so that you can conduct the study during tutorials (but all the participants can't be from the same group)
 - Study tasks. At least two concrete tasks should be evaluated.
 - Pilot studies and ethical considerations (including required ethics documents). At least prepare the consent form from [USyd - HREC](#).
 - Study plan including planned activities, data to be collected, data recording formats, and analysis plan.
2. Conduct a Think Aloud study. Groups of 4 must have 2 participants, whereas groups of 3 must have at least 1 participant. **- 5pts**
3. Outline key findings as successful and unsuccessful tasks (in a table). Use these results to discuss how LLM performed in terms of assuring usability in the web interface, and the usability considerations (max 200 words). **- 5pts**

Task 4 - (Research) LLM Security Awareness Evaluation - Task total 60pts

This task evaluates how well modern LLMs understand security concepts, assessing their biases and weaknesses in security-related advice. This is an open-ended **mini research** component, where you are encouraged to uncover novel insights about LLM security awareness.

Submission Requirement:

- A written report documenting your approach, key findings, and analysis.
- Focus on identifying **unexpected** or **previously unknown** behaviors in LLM responses.

Grading Criteria:

- Clearly explain your findings (**including system and threat models**). - 7.5pts
- Support your findings with quantitative data. - 7.5pts
- Compare your findings to existing research or common beliefs. - 7.5pts
- Demonstrate the novelty of your finding (Please read this blog post:). - 7.5pts

Each investigation area is graded out of **30 points**, with a maximum of **60 points** for two distinct investigation areas.

Suggested Investigation Areas:

- **Security Q&A with LLMs:** Evaluate LLM responses to security-related questions (e.g., secure coding, best practices). Assess their accuracy, completeness, and potential misinformation. Compare responses across models (e.g., ChatGPT, GPT-4, Google Bard, LLaMA-2).
- **Bias or Inconsistencies:** Investigate if LLMs consistently recommend secure practices or exhibit biases based on question wording, user persona, or context.
- **Adversarial Prompts / Jailbreaking:** Test whether LLMs can be manipulated into providing insecure responses. Explore adversarial prompts to assess their resilience to social engineering, prompt injection, or security policy bypasses.

Important:

- **Research novelty is key** – routine observations will not receive full marks.
- Keep explanations clear and concise.
- Cite relevant prior research where applicable.

Task 5 - Live Demo during Week 12 tutorial**- Task total 100pts**

- Demo will be 7 mins per group, followed by 3 mins Q&A. Every group member must present during the demo. The following points should be considered:
 - Introduction to the project and goals. **- Task total 10pts**
 - Live demo of the system. **- Task total 40pts**
 - Demonstrating/presenting security issues discovered and solutions. **- Task total 30pts**
 - Good response to Q&A. **- Task total 10pts**
 - Good time management. **- Task total 10pts**

Weekly Diary - All phases

You are required to submit a diary at the end of each project phase, indicating the individual contributions you made to the project and the report. This should be a summary of your activities for each week. You should follow the following tabular format.

Group Name		
Week	Student Name	Activities
W03	Jane Doh	Joined the meeting to discuss ideas and contributed to ideation
W03	John Doh	Joined the meeting to discuss ideas and contributed to ideation
W04	Jane Doh	Drafted project scope
W04	John Doh	Made the listed of planned activities
W05	...	...